LAUNCH THE PROJECT - 04

Joystick Controlled Robotic Arm

(Intermediate level Project)

On AYNOP® UNO Launchpad Kit



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1. Overview

This project demonstrates how to control a **servo motor like a robotic arm using a joystick** with the Arduino® UNO R4 Minima.

The joystick module provides **analog input**, which allows smooth control of the servo motor position.

In this project:

- Moving the joystick Up or Down controls the position of the servo motor.
- The servo behaves like a robotic arm joint, allowing smooth positioning.
- When the joystick returns to the center position, the servo **holds its current position**

You will learn to:

- Read **analog input** from a joystick module
- Control a **servo motor** using Arduino
- Implement a **dead-zone** to prevent unwanted movement
- Map user input into **precise mechanical motion**

This project introduces the concept of **manual robotic arm positioning**, commonly used in robotics, automation systems, and remote-controlled devices.

2. Components Required

- Arduino® UNO R4 Minima board
- USB Type-C data cable
- Joystick module
- SG90 servo motor
- 100 µF electrolytic capacitor (recommended for power stabilization)
- Jumper wires
- Breadboard

Note:
The capacitor is **recommended but not mandatory**. It helps stabilize the power supply when the servo motor draws sudden current.

3. Software Required

- Arduino IDE (v2.3.7 or later)
- Servo library (built into Arduino IDE)

  💡 **Tips**: No external installation needed. The Servo library comes pre-installed with Arduino IDE

Note:
We assume the **Arduino UNO R4 Minima board package** is already installed on your machine, as explained in the 00_Getting_Started/00_GettingStarted_Arduino_R4_Minima guide. If it is not installed, please refer to that document and complete the installation before proceeding.

4. Hardware Setup

This section explains how to connect the joystick module and servo motor.

4.1 Wiring Diagram

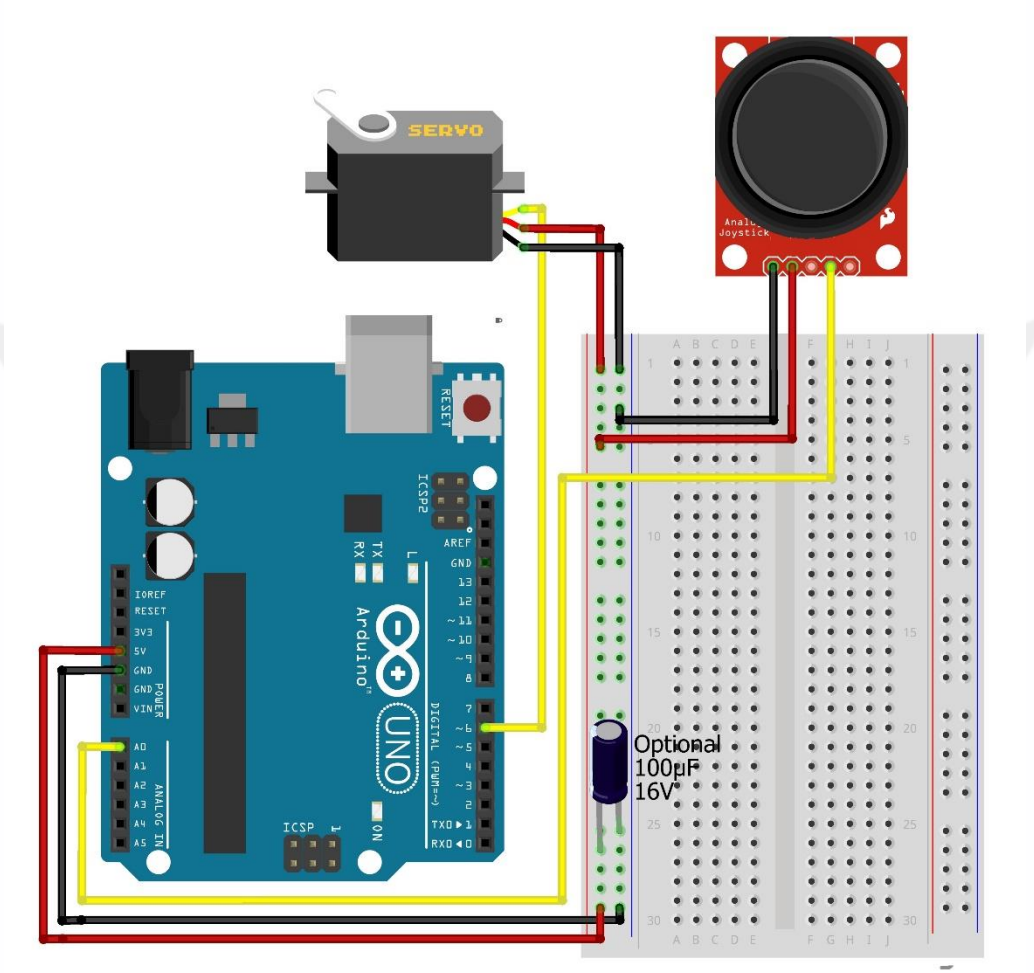


Figure 4.1 – Wiring diagram for Joystick Controlled Robotic Arm

Tip: Always disconnect the USB cable before making or changing hardware connections.

5. Principle – How It Works

The joystick module acts as an **analog input device** that allows the user to control movement manually. Inside the joystick, there are **two variable resistors (X and Y axes)**.
In this project, only the **Y-axis** is used.
When the joystick is moved:

- UP → Output voltage decreases
- Down → Output voltage increases

The Arduino reads this voltage using: **analogRead() → values from 0 to 1023**

Servo Control Logic

The servo motor represents a **robotic arm joint.** Instead of directly mapping the joystick value to the servo angle, this project uses **incremental control**

- If joystick moves UP → servo angle increases
- If joystick moves DOWN → servo angle decreases

This creates smooth and controlled movement, similar to how real robotic arms are operated.

Dead-Zone Concept

A small range around the joystick's center position is defined as a **dead-zone**.

When the joystick is in this range:

- No movement is applied to the servo
- The robotic arm holds its position

This prevents unwanted movement caused by small signal variations or noise.

6. Procedure – Steps to Run

1. **Build the Circuit**
   a. Connect the Joystick and servo motor as shown in the wiring diagram (Figure 4.1).

2. **Connect the Board**
   a. Use a USB Type-C data cable to connect your UNO R4 Minima to your computer.

3. **Open the Project Code**
   a. Simply **double-click** the file *04_Joystick_Controlled_Robotic_Arm.ino* in the project folder, and it will open directly in the Arduino IDE (if installed).

4. **Confirm Board Selection**
   a. The IDE usually auto-detects the UNO R4 Minima if the package is installed.
   b. If not installed, refer to the *00_Getting_Started/00_GettingStarted_Arduino_R4_Minima* document to install the necessary board package.
   c. Verify that *Arduino UNO R4 Minima* is selected in the IDE's board selector (top toolbar).

5. **Upload the Code**
   a. Click the **Upload** button (arrow icon) in the top-left corner of the IDE.
   b. Wait until the console displays "**Done uploading**."

6. **Observe the Behaviour**
   a. The servo motor starts at the **center position (90°)**.
   b. Move the joystick **Up or Down**.
   c. The servo motor rotates smoothly according to the joystick movement.
   d. When the joystick is released to the center, the servo **holds its current position.**

7. Expected Output

- **Joystick at Center:**
  - Servo holds its current position.
  - No movement occurs due to the dead-zone.

- **Move Joystick UP:**
  - Servo rotates gradually toward **0°**

- **Move Joystick Down:**
  - Servo rotates gradually toward **180°**.

- **Return Joystick to Center:**
  - Servo stops moving.
  - Holds its last position

💡 **Tip:** Move the joystick UP or Down and then release it. Observe how the servo holds its position like a **real robotic arm joint**.

8. Code

The source code for this project is included in the downloaded folder:

📂 uno-launchpad-kit/02_Intermediate_Projects/ 04_Joystick_Controlled_Robotic_Arm/ 04_Joystick_Controlled_Robotic_Arm.ino

👉 **Tip:**

- To open the project, simply **double-click the .ino file**. If the Arduino IDE is installed, it will launch automatically and load the code.
- If you **haven't installed the Arduino IDE yet**, please refer to:
  📂 uno-launchpad-kit/00_Getting_Started/00_GettingStarted_Arduino_R4_Minima
  to **download and install it.**

8.1 Function References

- setup() – Runs once when the board is powered on or reset. Used to initialize sensors, LCD, and I/O pins.
- loop() – Runs continuously after setup () finishes. Reads sensor values and updates the LCD at fixed intervals.
- analogRead(pin) – Reads the analog voltage from the joystick module.
- Servo.attach(pin) – Connects the servo object to a specific control pin.
- Servo.write(angle) – Moves the servo motor to a specified angle.
- constrain(value, min, max) – Limits a value within a specified range.
- delay(ms) – Adds a small delay for smooth servo motion.

📚 **For more details and advanced usage, visit:**
🔗 Arduino Language Reference — The official guide for all Arduino functions.

9. Troubleshooting Tips

- **Servo motor jittering?**
  - Add a **100 µF capacitor between 5V and GND** near the servo to stabilize the power supply.
  - Ensure all connections are tight and wires are not loose.

- **Servo not moving?**
  - Check wiring as per wiring diagram.
  - Ensure the servo is properly connected to the breadboard.

- **Servo moving randomly without joystick input?**
  - This may be due to noise around the center position.
  - Ensure the joystick is stable and properly connected.
  - Verify that the dead-zone values in the code are correct

- **Joystick not responding?**
  - Check wiring: VCC → 5V, GND → GND, VRy (Y-axis) → A0.
  - Ensure correct axis (Y-axis) is used in the code.

- **Servo not reaching full range (0° to 180°)?**
  - Move the joystick fully Up and down.
  - Some servos may have slightly limited mechanical range.

- **Arduino resetting when servo moves?**
  - This is due to power fluctuations.
  - Add a **100 µF capacitor** across 5V and GND.
  - Avoid powering multiple motors directly from the board

- **Still not working?**
  - Recheck all wiring connections carefully.
  - Disconnect power and rebuild the circuit step-by-step.
  - Test each module (Joystick & Servo motor) individually with help of basic projects.

10. License

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11. Support & Feedback

We value your feedback and are happy to assist with any questions, troubleshooting, or suggestions you may have.

Email: *support@aynop.com*

When sending an email, please include your kit name (AYNOP® UNO Launchpad Kit) and, if applicable, the project name in the subject line. This will help our team respond faster and more accurately. We aim to respond to all queries within 2–3 business days. Your feedback helps us improve our products and create even better learning experiences for all Arduino beginners.